

The background image shows several women working in a tea plantation. They are wearing headbands and carrying large, round, woven baskets on their heads, which are filled with harvested tea leaves. The women are dressed in traditional Indian attire, including sarees. The tea plants are lush green and arranged in neat rows. The overall scene is a vibrant green, with the women's colorful clothing providing a contrast.

LOW-COST LIGHTWEIGHT TEA LEAF COLLECTION BUCKET

A simple ergonomic design for sustainable and productive
harvesting

Team
Ergonomics

E/21/037 | E/21/094 | E/21/365

A LOW-COST ERGONOMIC SYSTEM FOR SUSTAINABLE HARVESTING

NEED

To design, build, and test a ventilated and ergonomically structured tea leaf collection bucket to study the mechanisms of heat reduction and load distribution.

AIM

To improve harvesting productivity of tea leaves and worker comfort by replacing currently available baskets with an optimized, low-cost solution.

Focus Area	Limitations of Existing Basket	Proposed Ergonomic Bucket
Air Flow	Heat develops inside the bucket, causing fermentation and permutations of tea leaves.	Reduce pressure development by applying support bars and producing targeted ventilation.
Capacity	Maximum collectable weight is limited to 6-7 kg.	Aim to increase the capacity to handle 10-11 kg safely.
Leaf Quality	Due to heating and pressure development, leaves at the bottom get crushed.	Rigid support bars maintain structural integrity, preventing crushing of bottom leaves.
Ergonomics	Causes shoulder pain in long term usage due to lack of anatomical support & Back Pain.	Apply specialized support padding and structural straps in the shoulder area.
Cost	Total cost for a currently available basket is Rs. 1275.	Aims to reduce manufacturing cost to Rs. 1000.

COMPARISON OF TEA LEAVES COLLECTION BUCKETS

Focus Area	Traditional Woven Baskets	Existing Alternative Baskets	Proposed Ergonomic Bucket	
				
	Air Flow	Suffer from limited ventilation.	Heat develops inside the bucket, causing fermentation and permutations of the tea leaves.	Features engineered targeted ventilation and a ventilated cavity to prevent internal heat buildup.
	Capacity	Holds approximately 10–11 kg when full, but has a heavy empty weight of 2.5–3 kg.	Maximum collectable weight is limited to only 6–7 kg.	Aims to safely handle 10–11 kg while using lightweight materials
	Leaf Quality	Limited ventilation can impact quality.	Leaves at the bottom get crushed due to heating and internal pressure development.	Rigid support bars maintain structural integrity, preventing the crushing and heat-induced permutation of bottom leaves
	Ergonomics	Carried with a strap around the head or shoulders, causing severe worker fatigue and neck strain.	Causes shoulder pain and back pain in long-term usage due to a lack of anatomical support.	Applies specialized support padding and integrated shoulder structural straps to eliminate localized strain and distribute loads
Cost	Inexpensive and widely used.	Currently available basket total cost is Rs. 1275, which limits widespread adoption.	Aims to reduce manufacturing cost to Rs. 1000, making it economically viable for quick recovery through productivity	

ENGINEERING DESIGN & METHODOLOGY



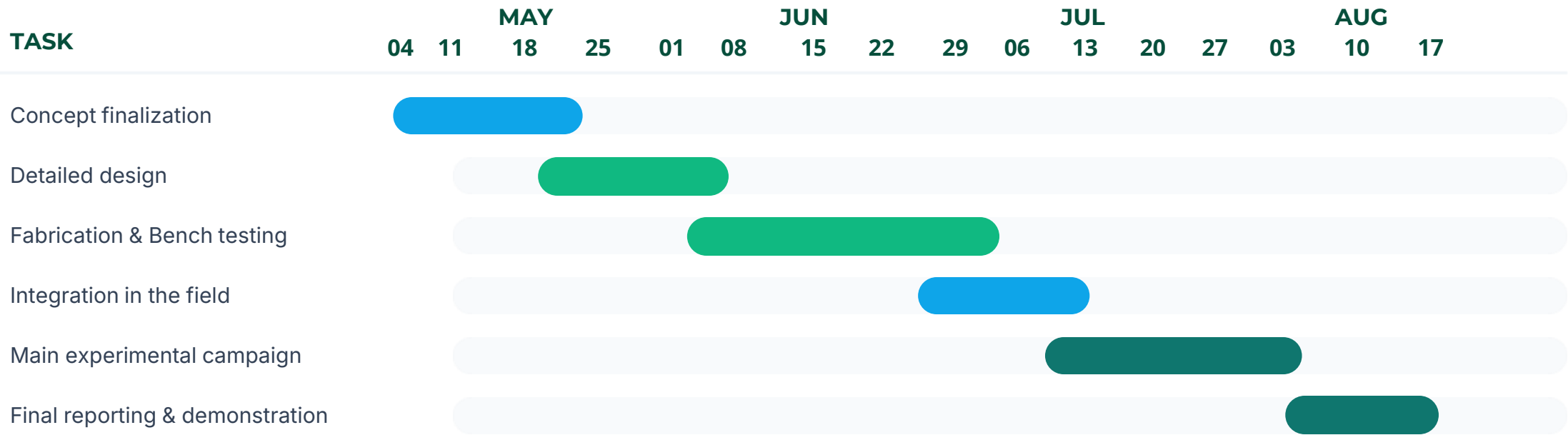
SOFTWARE TOOLS

- **SolidWorks** – CAD design of bucket frame and ventilation slots
- **ANSYS** – Thermal flow analysis and structural stress testing

METHODOLOGY

- Ergonomic baseline study using existing traditional baskets to identify pain points.
- CAD modeling of internal support frame with dominant load-bearing analysis.
- Selection of low-cost, lightweight synthetic materials to achieve the Rs. 1000 target.
- Field visualization and capacity testing to confirm the 11kg limit and zero leaf permutation.

MILESTONES & GANTT CHART (MAY 4 - AUG 17)



"BIG THREE" IMPACT



SOCIAL

Lowering physical strain directly supports local livelihoods. Eliminating chronic shoulder pain and fatigue improves the overall wellbeing, stability, and resilience of the plantation working communities.



ENVIRONMENTAL / QUALITY

Directly aligns with sustainable harvesting. Internal support bars prevent the crushing and heat-induced permutation of bottom leaves, drastically reducing crop waste and improving final tea quality.



ECONOMIC

Increasing capacity to 10-11 kg yields higher daily productivity. Lowering the manufacturing cost from Rs. 1275 to Rs. 1000 provides massive economic relief to plantation owners at a fleet scale.

THANK YOU



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